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Grouper Training - MCP AI - Lesson 00: Pre-work

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Hands on is optional. If you do not want to do it, you do not need to do anything here: just come to class and look on with someone else.

For people attending the AI MCP lesson in person who want to do the hands on. Bring a laptop with these steps done. They take about 30 to 45 minutes, most of it starting your GTE and connecting to it in step 7. If a step does not work, ask Claude Code first: paste in what you did and the error message. If you are still stuck, ask in the Slack channel, or come to class an hour early and an instructor will help.

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Context Protocol), so it can look up and change groups as the logged-in user.

1. Make sure you have GTE hours

You need at least 10 hours of GTE VM time for the class. Check your remaining hours on the GTE VM manager: it should show at least 10 hours. If it does not, ask in the Slack channel.

Do not use up your hours before the class starts. The test in step 7 only needs about an hour.

2. Get a Claude license

Claude Code needs a Claude license.

- **If you already have one** (Pro, Max, Education, or Enterprise), you can use it, as long as it is allowed to connect to arbitrary MCP servers. Some Education and Enterprise accounts do not allow this. Step 7 tests it: if your account cannot connect, get a Pro account
- **Otherwise**, sign up for a Pro account at https://claude.ai ($20, billed monthly). You only need it for one month: do not forget to cancel it if you do not want to keep it

3. Trust the GTE certificate

The GTE uses its own (self-signed) certificate for `https://localhost:8443`. Claude Code, the Claude app, and the app's built-in browser only connect to it if your computer trusts that certificate, so do this before you install them.

1. Download the attachment gte-localhost.crt to your `Downloads` folder. If Confluence says "Something went wrong. We couldn't generate a preview for this file.", ignore it and click the download button
2. Mac: open the "Terminal" app (in "Applications" → "Utilities") and run this. It asks for your password:

```
1  security add-trusted-cert -r trustRoot -k ~/Library/Keychains/log
```

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Root Certification Authorities", and finish

Every GTE uses this same certificate, and its private key is in the public GTE image. Trust it only for the training, and remove it afterward (see the end of this page).

If the GTE image is updated, its certificate changes. You can always get the current one from a running GTE, with its SSH tunnel open (Mac or Linux):

```
echo | openssl s_client -connect localhost:8443 2>/dev/null | opens
```

4. Install the Claude app and Claude Code

Install both. The command line is needed to add and log in to the Grouper MCP server. The app is an easier place to work, and its "Code" tab uses the same MCP servers.

1. Download and install the Claude app for Mac or Windows from https://claude.ai/download, and sign in with your Claude account
2. Install the Claude Code command line (below)

Mac or Linux, in a terminal:

```
curl -fsSL https://claude.ai/install.sh | bash
```

Windows, in PowerShell:

```
irm https://claude.ai/install.ps1 | iex
```

5. Approvals

Claude Code asks before it does things on your computer. Do **not** allow:

- Cookies or sign-ins from your web browser
- Folders outside `~/grouperMcpTraining`
- Access to other apps

Generally fine to allow:

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what they will change

6. Sign in and test

1. In a terminal, make the training folder, go into it, and run `claude`

   Mac or Linux:

   ```
   mkdir -p ~/grouperMcpTraining
   cd ~/grouperMcpTraining
   claude
   ```

   Windows PowerShell:

   ```
   mkdir $HOME\grouperMcpTraining -Force; cd $HOME\grouperMcpTrainir
   ```

2. Sign in with your Claude account
3. Ask it something simple and confirm you get an answer. Use the copy button to copy this prompt into Claude Code:

   ```
   what is Grouper?
   ```

4. Type `/exit` to quit

7. Connect Claude Code to a GTE

This checks that everything works before class, including that your Claude account can connect to an MCP server.

1. Start your GTE VM for 1 hour only, and connect to it with SSH, as described in Connecting to the Grouper Training Environment VM. Leave that SSH window open: Claude Code reaches Grouper through its tunnel. If the SSH window shows errors like `setsockopt TCP_NODELAY: Invalid argument`, press Enter and ignore them
2. Confirm https://localhost:8443/grouper loads in your browser (accept the security warning)
3. Download the Grouper v7 source code into the training folder, in a folder named grouper (about 225 MB unzipped). In class, Claude Code can read it

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Mac or Linux:

```
.zip && unzip -q grouper.zip && mv grouper-GROUPER_RELEASE_7.9.0 grouper
```

Windows PowerShell (unzipping can take several minutes):

```
cd $HOME\grouperMcpTraining; Invoke-WebRequest -Uri https://github.
```

4. Add the GTE MCP server for your user, so it works in every folder and in the Claude app Code tab:

```
claude mcp add --transport http -s user grouper-gte-local https://loca
```

Start Claude Code in the training folder. Mac or Linux:

```
cd ~/grouperMcpTraining
claude
```

Windows PowerShell:

```
cd $HOME\grouperMcpTraining; claude
```

The first time, Claude Code asks you to sign in with your Claude account, then asks if you trust this folder. Sign in, and choose yes for the folder.

5. In Claude Code, type `/mcp`, select "grouper-gte-local", and select "Authenticate"
6. A browser opens. Accept the security warning, log in as `banderson` / `password`, and approve read-only access
7. Back in Claude Code, use the copy button to copy this prompt:

   ```
   what groups is banderson a member of in Grouper?
   ```

   You should see a list of groups from the GTE
8. Make Claude always ask before it changes Grouper. Use the copy button to copy this prompt into Claude Code in the same conversation:

   ```
   Add this to my user memory, ~/.claude/CLAUDE.md,
   so it applies in every folder: before any write
   operation with the grouper-gte-local MCP server
   ```

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```
7  what you will change and wait for my approval.
```

9. Let the read-only Grouper tools run without asking each time. Write tools still ask. Use the copy button to copy this prompt into Claude Code in the same conversation:

```
1  In my user settings, ~/.claude/settings.json, add
2  the read-only grouper-gte-local MCP tools to the
3  permissions allow list, so they run without asking
4  me: doc_search, attribute_def_name_find,
5  group_find, folder_find, attribute_assignment_get,
6  audit_get, privilege_get, entity_get_groups,
7  group_get_members, memberships_get, entity_get,
8  group_has_member, recipe, sql_get_schema,
9  sql_select, admin_external_system_get,
10 admin_daemon_job_message, admin_daemon_logs,
11 admin_config_search, admin_daemon_names, and ldap.
12 Tool names look like
13 mcp__grouper-gte-local__group_find.  Do not add
14 any write tools.  Show me the result.
```

10. Type `/exit` to quit Claude Code. If the Claude app is open, quit it completely (Mac: Cmd+Q; Windows: right-click it in the taskbar and close it) so it picks up the MCP server and settings when it opens again

Then check the Claude app, since most of the class works there:

You do all MCP setup and authentication in Claude Code on the command line. The Claude app uses the same MCP servers, even if it does not list `grouper-gte-local` with its connectors.

11. Open the Claude app and click "Code" at the top
12. Start a new chat. In the row under the prompt box, click the folder control, click "Open folder...", and choose the `grouperMcpTraining` folder in your home folder. If the app asks to save it as an environment, say yes
13. In the model picker, select Opus 5.5 and "Medium" effort. If you set the model, effort, and folder first thing in a new conversation, they stick for your new conversations.

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You do not need "Accept edits": that only skips the question for file edits

15. Ask the same question. Use the copy button to copy this prompt into the Claude app:

```
what groups is banderson a member of in Grouper?
```

You should see the same list. If it cannot reach `grouper-gte-local`, ask in the Slack channel

If the Claude app shows an error that `grouper-gte-local` could not start for Cowork and Code sessions, it has its own copy of the server in its config file, often added by an earlier setup prompt, and that copy cannot sign in to Grouper. The Code tab does not need it: it uses the server you added with `claude mcp add`. Use the copy button to copy this prompt into Claude Code:

```
Check the Claude app config file,
claude_desktop_config.json.  If it has a
grouper-gte-local MCP server, remove it, and show
me the result.  Do not change my Claude Code MCP
servers.
```

Then quit the Claude app completely and open it again.

When it works, terminate your VM in the GTE VM manager so you keep your hours for class. If you cannot get this working, come to class an hour early and an instructor will help.

On the day of the training

A Claude Pro plan has a usage limit that resets every 5 hours, counted from your first message. The Claude app and Claude Code share the same limit. Time it so your limit resets during the class. All times are Central time (Minneapolis time) on Monday, October 26.

Time	What to do

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	window would start your 5 hours early, and your limit would not reset during the class
10am	Start Claude Code and send it any prompt, e.g. hello. This starts your 5 hours
3pm	Your limit resets, during the class, so you are less likely to run out during the hands on

After the training: keep experimenting

On your own time, you may run out of your 10 GTE hours in AWS. You can also run the GTE on your own computer, without the AWS VM:

1. Install Podman Desktop or Docker Desktop
2. Follow Run locally, on the Grouper Training Environment page

The GTE is then at https://localhost:8443/grouper as before, so the same MCP setup works. If your local GTE has a different certificate, get it from the running GTE with the command at the end of step 3, and trust it the same way.

After the training: remove the certificate

Mac, in the "Terminal" app:

```
FINGERPRINT=$(openssl x509 -in ~/Downloads/gte-localhost.crt -noout
security delete-certificate -Z $FINGERPRINT ~/Library/Keychains/log
```

Windows: open "Manage user certificates", go to "Trusted Root Certification Authorities" → "Certificates", and delete the one issued to `localhost`.

After the training: remove the GTE MCP server

When you no longer use the GTE, remove its MCP server. In a terminal:

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If it says the server is not found, it was saved only for the training folder: run the same command without `-s user` from `~/grouperMcpTraining`. Then clean up the settings from step 7. Start `claude` and use the copy button to copy this prompt into Claude Code:

```
Remove the grouper-gte-local MCP tools from the
permissions allow list in ~/.claude/settings.json,
and remove the grouper-gte-local approval rule
from ~/.claude/CLAUDE.md.  Show me the result.
```

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